

Making OpenAD Robust to Query Phrasing in 3D Affordance Detection

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Abstract—Open-vocabulary affordance detection lets a robot or an assistive interface ask, in free-form language, where on a 3D object an action can be performed. OpenAD realises this by matching per-point PointNet++ features against CLIP text embeddings of arbitrary query strings. It is trained with single-word labels, which raises the question of how far its open-vocabulary ability extends to the phrasing that people naturally use. We answer it with a chain of connected experiments. We first reproduce the released model on the 3D AffordanceNet validation split (14.40 mIoU against 14.37 reported) and use the resulting evaluation harness to measure the effect of wording: rewording five affordances as questions or descriptions changes the full-vocabulary mIoU from about 41 to about 30, and the score maps stay spatially correlated across phrasings. This locates the opportunity in the alignment between point features and sentence-style text. We therefore introduce prompt-augmented fine-tuning, which keeps the architecture, the CLIP text encoder and the point labels unchanged and samples paraphrases of the affordance names at every training step. Updating 101,377 parameters, under six percent of the network, raises mIoU on held-out phrasings that were fixed before training by 11.2 points for questions and 10.5 points for descriptions, which recovers more than 90 percent of the difference, while the score obtained with the original words is preserved. A capacity study over three unfreezing depths (67,073, 101,377 and 743,041 parameters) shows that the gain is already present with the alignment head alone and remains at the same level as more of the decoder is released, and two query-independent floors place every score on the metric scale. A qualitative study and an interactive prototype complete the work. The results show that phrasing robustness can be added to OpenAD at low cost.

Index Terms—affordance detection, open vocabulary, point clouds, CLIP, prompt robustness, fine-tuning

I. INTRODUCTION

An affordance is an action that an object offers to an agent. For example, a mug can be grasped by its handle, a chair can be sat on, a knife cuts with its blade [1]. For a robot or an assistive interface, knowing *where* on an object an action can be performed is more useful than knowing only what the object is. This has led to per-point affordance detection on 3D point clouds, where every point of an object receives an affordance label [2]. Classical detectors of this kind predict from a fixed list of affordance classes, so a new action requires new annotations and a new classifier.

OpenAD [3] removes the fixed list. Instead of a classification layer, it embeds the query words with a frozen CLIP

text encoder [4] and labels each point with the query whose embedding is closest to the point feature. In principle any word can be used at test time. In practice a user or a robot planner does not type the exact words that were used for training. A user may write “Where can I grasp this object?” or “the part used for holding” where the model saw only “grasp”. Whether an open-vocabulary detector trained on single words copes with such phrasing is an empirical question, and the answer determines whether the model can be placed behind a natural-language interface at all.

This report investigates that question with the released OpenAD PointNet++ model and the 3D AffordanceNet data, in a chain of connected experiments in which each result determines the next. We ask five questions.

- 1) Can we reproduce the published closed-set and open-vocabulary numbers with the released checkpoint and our own evaluation code?
- 2) How much does accuracy change when the training words of some affordances are replaced by questions or descriptions with the same meaning?
- 3) Can a lightweight adaptation that changes only the language supervision restore accuracy, without touching the point backbone, the text encoder or the ground-truth masks?
- 4) Does the restored accuracy transfer to phrasings that were never used for training?
- 5) How many parameters need to change, and where in the network does the improvement reside?

Our contributions are the following. (i) An evaluation harness that scores any set of query strings against the same data, metric and decision rule, validated against the official evaluator and by unit tests, together with a reproduction of the published OpenAD numbers and two query-independent floors that anchor the metric scale. (ii) A quantitative characterisation of how query phrasing changes the pretrained model, with a per-class decomposition and a score-map analysis that identifies where the opportunity lies. (iii) Prompt-augmented fine-tuning of the alignment head and the last feature-propagation layer (101,377 parameters), evaluated on phrasings that were frozen before training. (iv) A capacity study over three unfreezing depths (67,073, 101,377 and 743,041 parameters) that shows

how little of the network the improvement requires. (v) A qualitative analysis of representative cases and an interactive prototype built around the same model.

In brief, the wording of the query moves the released model by about 12 mIoU points when five affordances are phrased naturally, and a fine-tune of 101,377 parameters with paraphrase supervision recovers more than 90 percent of that difference on phrasings it has never seen, without changing the architecture, the text encoder or the labels. Releasing more than seven times as many parameters gives the same level of performance, so the alignment head is where the improvement is made.

The scope is deliberately focused, which keeps every result attributable. All experiments start from the weights released by the OpenAD authors. The architecture, the CLIP text encoder, the per-point labels and the loss are unchanged, and only the language supervision differs. We propose no new architecture; the contribution is the measurement and a lightweight remedy. Numbers are labelled as “cited” when taken from the OpenAD paper and as “ours” when produced by the code in our repository¹.

II. RELATED WORK

Affordance detection in 3D. 3D AffordanceNet [2] provides a benchmark of point clouds with per-point affordance labels and evaluates closed-set networks in full-shape and partial-view settings. Other work predicts affordance heat maps from object-object interaction [5], from 2D interaction images [6], or from natural-language queries [7]. OpenAD [3] introduced the open-vocabulary version of the task on point clouds by learning a shared space for point features and text. Follow-up work adds knowledge distillation and a text-point correlation module to this setting [8]; here we study instead how the released OpenAD model responds to the wording of the query.

Language-driven point cloud understanding. CLIP [4] aligns images and text at web scale. PointCLIP [9] projects point clouds to depth images to reuse CLIP for classification, and OpenScene [10] distills CLIP features into a 3D network for open-vocabulary scene understanding. OpenAD differs in that its point network is trained directly against frozen CLIP text embeddings, so no image branch is needed.

Point networks. PointNet [11] processes unordered points with shared per-point MLPs and a symmetric pooling function. PointNet++ [12] adds a hierarchy of local neighbourhoods obtained by farthest point sampling and ball queries, and dynamic graph convolution [13] is an alternative that OpenAD also supports. We use the PointNet++ variant.

Prompt sensitivity and adaptation. It is known that the wording of a text prompt changes the behaviour of CLIP-based classifiers. The original CLIP work uses hand-designed templates and prompt ensembles [4], and CoOp [14] learns continuous prompt vectors. Our approach differs in that we

keep the text encoder and the text fixed, and instead expose the point-side alignment head to many paraphrases during training. Updating few parameters follows the observation that full fine-tuning can distort pretrained features [15], and is in the spirit of lightweight adaptation of CLIP through feature adapters [16] and of robust fine-tuning of zero-shot models [17].

III. DATASET AND EVALUATION PROTOCOL

A. Dataset

We use the full-shape setting of 3D AffordanceNet [2] as distributed with OpenAD. According to the OpenAD paper [3], the dataset contains 22,949 object instances from 23 object categories and 18 affordance classes. Each object is a point cloud of $N = 2048$ points, as fixed in the OpenAD paper, and each point carries one integer label in the released files, either one of 18 affordances or a background class `none`, giving 19 classes in total. The affordance names, in label order, are `grasp`, `contain`, `lift`, `openable`, `layable`, `sittable`, `support`, `wrap_grasp`, `pourable`, `move`, `display`, `pushable`, `pull`, `listen`, `wear`, `press`, `cut`, `stab` and `none`. We rely on this order matching the dataset labels, which we verified on the data.

The released files contain a training and a validation split and no separate test split. As in the released OpenAD code, we report results on the validation split. It contains 2,285 objects. The categories are strongly imbalanced. The Table (799 objects) and Chair (611) categories account for 62% of the objects, StorageFurniture has 218, and the smallest categories are Scissors (6), Bag (12), Keyboard (15) and Dishwasher (16).

B. Preprocessing

Points are centred on their centroid and divided by the largest distance to the centroid, so that each object lies in the unit sphere. This is the normalisation of the released data loader, and our inference code repeats it for arbitrary input clouds. We apply no geometric augmentation. The only augmentation in this work acts on the language side (Section IV-E).

C. Label protocol and open-vocabulary test

Let $\mathcal{Q} = (q_1, \dots, q_K)$ be a list of query strings aligned with the class indices. In closed-set evaluation, $\mathcal{Q}$ contains the 19 training words. OpenAD tests open-vocabulary behaviour by replacing the 18 affordance words with synonyms while keeping the labels unchanged. The synonyms are `grab`, `accommodate`, `raise`, `unlock`, `rest`, `take a seat`, `bear`, `wrap`, `pour`, `reposition`, `demonstrate`, `push`, `drag`, `hear`, `clothe`, `thumb`, `slice` and `jab`. The class `none` is kept. The released configuration that we run evaluates exactly these 19 queries, and the close match between our reproduction and the number reported in the OpenAD paper (Section VI-A) indicates that it is the benchmark that the paper describes.

¹<https://github.com/mehreen019/Open-Vocabulary-3D-Affordance-Detection>

D. Metrics

The predicted class of point i is the index of the query with the highest score, $\hat{y}_i = \arg \max_k S_{ik}$ (Section IV). Intersection, union and support are accumulated over all points of all objects in the split. For class c , with the bracket equal to 1 if the condition holds and 0 otherwise,

$$\text{IoU}_c = \frac{\sum_i [\hat{y}_i = c \wedge y_i = c]}{\sum_i [\hat{y}_i = c \vee y_i = c] + \epsilon}, \quad \epsilon = 10^{-6}, \quad (1)$$

and $\text{mIoU} = \frac{1}{19} \sum_c \text{IoU}_c$, which includes the background class. We also report point accuracy and mean class accuracy (mean per-class recall). These definitions follow the official evaluator, which we compared line by line with `utils/eval.py` of the OpenAD repository, and our implementation is tested against hand-computed confusion matrices. All IoU and mIoU values in this report are given in percent, i.e. 100 times the quantities of Eq. (1).

E. Phrasing conditions

A phrasing condition changes only the strings in $\mathcal{Q}$. We studied five affordances, namely grasp, contain, sittable, pourable and cut. For each we wrote phrases in three forms, namely a *label* (a single word), a *question*, and a *description*. The phrases are split into two disjoint groups, *fine-tuning* phrases (7 per affordance, 35 in total) and *held-out* phrases (4 per affordance, 20 in total). The split was written into a version-controlled configuration file before any fine-tuning was run, and a unit test fails if a string occurs in both groups. In a condition such as “question, held-out”, the five studied affordances are replaced by their held-out question and the other 14 classes keep their canonical word. The labels of the points do not change. Table I lists the phrases used for evaluation. The held-out label forms are defined in the configuration but are not part of the comparison in Section VI.

Held-out phrases differ from the training phrases as strings, not necessarily as meanings. They are new wordings of affordances that the model already knows, so the experiment measures generalisation across wording and not across affordances.

F. Evaluation noise

The point network selects its centroids by farthest point sampling from a random start, so repeated evaluations of one checkpoint differ slightly. We used this to calibrate the harness. Across five evaluations of the pretrained checkpoint with canonical words we obtained 41.19, 41.25, 41.28, 41.44 and 41.53 mIoU (mean 41.34, range 0.34). In the held-out question condition four evaluations gave 29.22, 29.27, 29.29 and 29.63 (range 0.41), and in the held-out description condition 30.22, 30.23, 30.28 and 30.40 (range 0.18). The harness therefore resolves differences of about 0.4 mIoU, and we use this value as the yardstick for every comparison below, treating differences under about 0.5 mIoU as equivalent. Each table entry is one evaluation pass, and the pretrained rows of Tables IV, V and VII are the passes of the respective comparison runs.

IV. METHOD

A. Task formulation

The input is a point cloud $P \in \mathbb{R}^{N \times 3}$ and a list of K query strings $\mathcal{Q}$. The output is a matrix $S \in [0, 1]^{N \times K}$ whose row i is a probability distribution over the queries for point i . Two uses occur in this work. For evaluation, $\mathcal{Q}$ holds all 19 class strings and the prediction is the arg-max. For a single user query, $\mathcal{Q} = (q, \text{none})$ and the score of the query, S_{i1} , is used as a heat map. Because of the softmax, a score is always relative to the competing queries. The same phrase can receive different scores depending on the other strings in $\mathcal{Q}$, which matters for interpreting the maps in Section VII.

B. Architecture

Figure 1 shows the model, which we did not modify. It has three parts.

Point encoder. A PointNet++ network [12] with multi-scale grouping. Set-abstraction layer 1 samples 512 centroids by farthest point sampling and gathers neighbours in balls of radius 0.1, 0.2 and 0.4 (32, 64 and 128 neighbours), applying a shared MLP per scale. Layer 2 samples 128 centroids with radii 0.4 and 0.8 (64 and 128 neighbours), and layer 3 pools all points into one global feature of 1024 dimensions. Three feature-propagation layers then upsample back to N points by inverse-distance interpolation over the three nearest neighbours, concatenated with skip connections, and produce a 128-dimensional feature per point. A 1×1 convolution from 128 to 512 channels followed by batch normalisation (the *alignment head*) gives the point embedding $f_i \in \mathbb{R}^{512}$. Without the frozen text encoder, the point network has 1,782,765 parameters, which we counted by instantiating the model.

Text encoder. The ViT-B/32 text transformer of CLIP [4], frozen, maps each query string to $t_k \in \mathbb{R}^{512}$. Its parameters are excluded from every optimiser and its outputs are computed without gradients.

Alignment. Point and text embeddings are compared by cosine similarity, scaled by a learnable scalar s and normalised over the queries,

$$z_{ik} = s \frac{f_i^\top t_k}{\|f_i\| \|t_k\|}, \quad S_{ik} = \frac{\exp(z_{ik})}{\sum_{j=1}^K \exp(z_{ij})}. \quad (2)$$

In the released code s is initialised to $\ln(1/0.07) \approx 2.659$ and is used directly, not exponentiated as in CLIP, so the initial temperature is about 0.38. Reading the released checkpoint shows that training has increased s to 25.41, an effective temperature of about 0.039. Fine-tuning moves it to 25.67.

C. Objective and original training recipe

Training uses a class-weighted negative log-likelihood of the softmax output,

$$\mathcal{L} = - \sum_{i=1}^N w_{y_i} \log S_{i, y_i}, \quad w_j = \left(\frac{\max_c n_c}{n_j} \right)^{1/3}, \quad (3)$$

TABLE I

QUERY STRINGS USED IN THE EVALUATION FOR THE FIVE STUDIED AFFORDANCES. “SEEN” PHRASES ARE MEMBERS OF THE FINE-TUNING SET AND “HELD-OUT” PHRASES WERE NEVER USED FOR TRAINING OR FOR CHOOSING AN EPOCH. THE REMAINING 14 CLASSES ALWAYS USE THEIR CANONICAL WORD.

Affordance	Canonical	Question, seen	Question, held-out	Description, seen	Description, held-out
grasp	grasp	Where can I grasp this object?	Where should I take hold of this?	the part used for holding	the surface meant for gripping by hand
contain	contain	Where can this object contain something?	Where does it keep things inside?	the hollow region that holds contents	the cavity that carries material inside
sittable	sittable	Where can I sit on this object?	Where would a person sit down here?	the surface you sit on	the part that supports a seated person
pourable	pourable	Where can I pour from this object?	Where does the liquid come out when tipped?	the opening liquid is poured from	the lip through which contents flow out
cut	cut	Where can I cut with this object?	Where is the part that cuts things?	the sharp edge used to cut	the honed edge intended for slicing

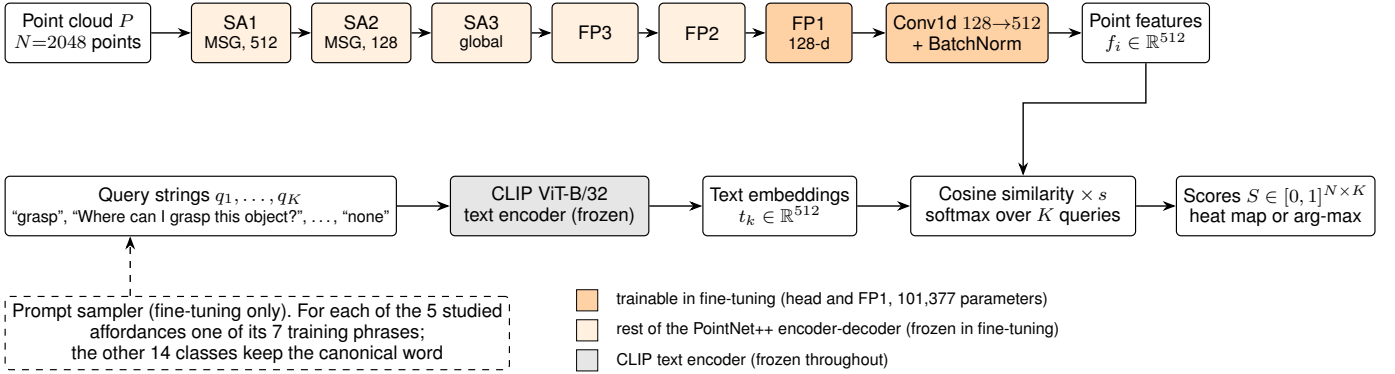


Fig. 1. Pipeline of OpenAD and of our prompt-augmented fine-tuning. Per-point features from a PointNet++ encoder-decoder are compared with CLIP embeddings of arbitrary query strings by cosine similarity, and a softmax over the queries gives per-point scores. The number of queries K is not fixed by the architecture. In fine-tuning only the alignment head is trained, and only the query strings change from step to step; the per-point labels are the same as in the original training.

where n_j is the number of training points of class j [3]. The weights compensate for the dominance of the background and of frequent affordances. The cube root softens the correction, because full inverse-frequency weights would give very rare classes a large gradient from few points. The released configuration trains for 200 epochs with Adam [18] at learning rate 10^{-3} , weight decay 10^{-4} , batch size 16, a step decay of the learning rate by a factor 0.5 every 20 epochs (floor 10^{-5}), and a matching decay of the batch-normalisation momentum [19]. We did not repeat this training; the released checkpoint is the starting point of everything below.

D. Why phrasing can matter

In training, the model only ever sees 19 distinct strings, so the alignment head has 19 target directions in the CLIP text space. CLIP embeds a full sentence and a single word differently, even when they describe the same affordance. A point feature aligned with the embedding of “grasp” need not be close to the embedding of “Where can I grasp this object?”. In the 19-way arg-max, the rephrased class then competes against 14 classes that still use the exact words seen in training. This is our working hypothesis for the mechanism, and the measurements in Section VI are consistent with it.

E. Prompt-augmented fine-tuning

The modification changes the language supervision and nothing else. At each training step, we draw a fresh query list $\hat{Q}$ in which, for each studied affordance a , we sample one phrase uniformly from its seven fine-tuning phrases (three labels, two questions, two descriptions), and all other classes keep their canonical word. The list is shared by the mini-batch. The point labels, the data, the loss of Eq. (3) and its class weights are those of the original training. The objective is therefore the expected loss over the phrase distribution, so the head is asked to place the points of an affordance close to the whole family of its paraphrases and not close to one word. Held-out phrases are never sampled; a unit test checks this.

What is trained. The text side is frozen, so the part of the network that can adapt to new wording lies at the end of the point branch. We therefore study a ladder of three nested configurations, each of which releases more of the decoder. (H) The alignment head (conv1, the affine parameters of bn1 and the scale s), which is $65,536 + 512 + 512 + 512 + 1 = 67,073$ parameters. (H+F) The head and the last feature-propagation layer fp1 (two 1×1 convolutions and two batch-normalisation affine pairs, 34,304 parameters), together 101,377 parameters, or 5.7% of the 1,782,765 parameters of the point network.

(D) The head and all three feature-propagation layers `fp1` to `fp3`, together 743,041 parameters, or 41.7%. A tensor-by-tensor comparison with the pretrained checkpoint confirms that exactly 5, 13 and 29 tensors differ in the three configurations (67,073, 101,377 and 743,041 elements) and that the batch-normalisation running statistics are unchanged. The set-abstraction encoder is frozen in all three, and the whole network is kept in evaluation mode so that batch-normalisation statistics do not drift. Confining the update to the end of the network has three motivations. First, the text side is frozen and unchanged, so the discrepancy that we want to remove lies in how point features are mapped into the text space, which is determined by the head and by the layers that produce its input. Second, a small update limits the risk of distorting features that the network learned in 200 epochs [15]. Third, it is cheap, since each epoch, including a full validation pass, took about 14 minutes on a laptop GPU.

Reported model. The three configurations are trained with the same data, phrases, loss, learning rate and seed and evaluated on the same conditions, so that differences between them are attributable to capacity alone (Section VI-E). We report head plus `fp1` as the main model because it reached the highest held-out scores of the ladder. As this choice used the held-out scores, its gains are best read together with those of the other two configurations, which lead to the same conclusions.

Optimisation. Adam with learning rate 10^{-4} (ten times below the original), no schedule, weight decay 10^{-4} , batch size 16 with shuffling, seed 1, and 8 epochs (5 epochs for the head-only configuration). These values were fixed in advance and not tuned. The lower rate is meant to adapt the network without undoing pretraining. After every epoch we compute the canonical-word mIoU on the validation split and keep the checkpoint with the highest value. This epoch selection uses canonical words only, so held-out phrasings play no role in choosing the checkpoint within a configuration. The selected checkpoint is that of epoch 4 in all three configurations.

Scope. Five of the 18 affordances are rephrased during training. The other classes see only their canonical word, so they act as the fixed competition against which the rephrased classes are evaluated.

F. Design decisions in brief

Table II summarises the main decisions of the pipeline, the alternative that each one excludes and the reason. The first five are properties of OpenAD that we inherit and the last four are ours. The reasons are design rationale, and the capacity ladder of Section VI-E tests the decision on which layers to train empirically.

V. EXPERIMENTAL SETUP AND BASELINES

Experimental design. The experiments are ordered so that each one supplies the input of the next. The reproduction validates the evaluation harness on which all later measurements rely (Section VI-A). The sensitivity study uses the harness to locate where the wording of a query matters and, through

TABLE II
DESIGN DECISIONS, THE ALTERNATIVE THAT EACH EXCLUDES, AND THE REASON.

Decision	Alternative	Reason
Frozen CLIP text encoder	Trainable text tower	Keeps the embedding geometry that relates synonyms; training sees only 19 strings, which a trainable encoder could fit without generalising
Cosine similarity with a learnable scale	Linear classifier	The number of classes is not fixed, and the scale sharpens a softmax that cosine values in $[-1, 1]$ would leave flat
Weighted NLL, cube-root weights	Unweighted cross-entropy	Background and frequent affordances dominate the point counts; the cube root limits the gradient of rare classes
FPS with multi-scale ball queries	Random sampling, single radius	Even coverage of the surface and robustness to non-uniform density
Adam, step-decayed rate	SGD	Published recipe of the released checkpoint, kept as is
Vocabulary of fixed length $K=19$ during fine-tuning	Paraphrases as extra classes	The softmax, the competition between classes and the metric stay exactly those of pretraining
Head and last propagation layer only	Full fine-tuning	The mismatch lies in the point-to-text mapping; few parameters limit forgetting and cost; a ladder of three configurations shows that the gain is already present with the head alone and stays at the same level when the whole decoder is released (Sec. VI-E)
Learning rate 10^{-4}	Original 10^{-3}	Adapt the network without undoing pretraining
Epoch selection on canonical validation mIoU	Selection on held-out phrasings	Keeps held-out phrasings out of the choice of epoch; the choice among the three layer configurations used them, and all three are reported (Sec. VI-E)

the score maps, where the opportunity lies. Prompt-augmented fine-tuning tests the resulting hypothesis with a minimal intervention and with held-out phrasings. The capacity ladder then asks how much of the network the intervention needs (Section VI-E). Two query-independent floors give scale to every score, and the qualitative study and the prototype show the effect on individual objects (Section VII).

Models compared. (B0) Two query-independent floors: a majority predictor that assigns every point to the most frequent class (`none`), and a predictor that draws a uniform random class for every point. Neither sees the point cloud or the query, and both anchor the scale of the metric. (B1) The pretrained OpenAD PointNet++ checkpoint for the full-shape setting released by the authors, loaded without change and pinned to the upstream commit `b082265`. (B2) Our prompt-augmented fine-tune of B1 that trains the alignment head and the last feature-propagation layer, which is the reported model. (B2', B2'') The same fine-tune with the head only and

with the whole feature-propagation decoder, which form the capacity ladder together with B2. In the reproduction table we also quote published open-vocabulary baselines (TZSLPC, 3DGenZ and ZSLPC, all with CLIP text encoders) from the OpenAD paper. They are marked as cited.

Fairness of the comparison. The pretrained and the fine-tuned model are evaluated in one script call, on the same validation objects, in the same order, with the same normalisation, the same metric implementation, the same arg-max decision rule and the same query lists. The only quantity that varies between conditions is the set of strings. The fine-tuning phrases, the held-out phrases and the vocabulary order are read from the same configuration files by both the training and the evaluation code. The training loop uses only the official training split. We trained three configurations (seed 1) and report all of them (Section VI-E). The learning rate and the number of epochs were fixed in advance and not searched.

Implementation. PyTorch 2.5.1 with CUDA 12.1, Python 3.9.8, OpenAI CLIP package, one NVIDIA RTX 4060 Laptop GPU. The reported fine-tuning took 6,834 s for 8 epochs (about 114 minutes), the head-only configuration 4,261 s for 5 epochs (about 71 minutes) and the decoder configuration 6,726 s for 8 epochs (about 112 minutes), each epoch including a full validation pass. The code contains 16 unit tests that run without a GPU and cover the metrics, the prompt sets and their leakage rules, the freezing logic and the application backend.

VI. RESULTS AND ANALYSIS

A. Reproduction of the pretrained model

Table III compares our evaluation of the released checkpoint with the values in the OpenAD paper. In the open-vocabulary setting we obtain 14.40 mIoU, 46.33% accuracy and 19.63% mean class accuracy against 14.37, 46.31 and 19.51 reported, so the evaluation pipeline and the checkpoint reproduce the published result. The closed-set mIoU of 41.25 is within 0.75 of the reported 42.00, and the closed-set point accuracy (68.51%) within 0.52 points of the reported 69.03%. The open-vocabulary numbers, on which the rest of the study builds, match the published ones to within 0.03 mIoU. The two query-independent floors of Table III place these scores on the scale of the metric. A majority predictor reaches 43.04% point accuracy but only 2.27 mIoU, because it scores zero IoU on all 18 affordances and the mIoU weights the 19 classes equally, and a random predictor obtains 1.47 mIoU and 5.28% accuracy, which is chance level for 19 classes. Every model in this report lies far above both floors, and the open-vocabulary score of 14.40 is already more than six times the majority floor.

The open-vocabulary mIoU is about three times lower than the closed-set one, and the per-class view in Fig. 2 shows where the difference comes from. Ten of the 18 synonym queries reach an IoU below 5% (for example grab for grasp at 1.7% and accommodate for contain at 0.2%), while slice for cut (55.0% against 54.7% with the canonical word) and hear for listen (49.5% against 52.1%) keep almost their full accuracy. Transfer to new words in this benchmark is thus

TABLE III
REPRODUCTION OF OPENAD (FULL-SHAPE, POINTNET++, CLIP) AND QUERY-INDEPENDENT FLOORS. “CITED” ROWS ARE TAKEN FROM THE OPENAD PAPER [3] AND WERE NOT RUN BY US. “OURS” ROWS ARE OUR EVALUATION OF THE RELEASED CHECKPOINT ON THE VALIDATION SPLIT (2,285 OBJECTS). THE FLOOR ROWS ARE COMPUTED WITH THE SAME METRIC CODE AND DO NOT DEPEND ON THE QUERY.

Setting	Method	mIoU	Acc	mAcc
Closed-set	OpenAD, cited	42.00	69.03	67.31
	OpenAD checkpoint, ours	41.25	68.51	66.73
Open-vocab.	TZSLPC, cited	3.86	42.97	10.37
	3DGenZ, cited	6.46	45.47	18.33
	ZSLPC, cited	9.97	40.13	18.70
	OpenAD, cited	14.37	46.31	19.51
	OpenAD checkpoint, ours	14.40	46.33	19.63
Floor	Majority class, ours	2.27	43.04	5.26
	Random class, ours	1.47	5.28	5.43

non-uniform: some word pairs transfer almost completely and others do not, which makes the benchmark a good testbed for asking what decides whether a phrase transfers. The next subsection isolates this for five affordances under controlled wording.

B. Effect of query phrasing on the pretrained model

Table IV evaluates the pretrained checkpoint with the five studied affordances rephrased as held-out questions or descriptions. The mIoU changes from 41.53 with canonical words to 29.63 for questions and 30.28 for descriptions. The change is concentrated in the rephrased classes: their mean IoU goes from 48.6% to 9.1% (questions) and 11.9% (descriptions), and grasp, contain and pourable drop below 0.5% for both forms. Since the metric is a mean over 19 classes, these five classes account for $(242.8 - 45.3)/19 = 10.4$ of the 11.9 mIoU points for questions and for $(242.8 - 59.6)/19 = 9.6$ of 11.3 for descriptions (sums of the columns of Table IV). The effect is therefore well localised, and the 14 classes that keep their canonical word account for the remaining 1.5 and 1.7 points, which probably reflect points that are reassigned to them from the rephrased classes. Cut retains a large part of its IoU (55.2% to 43.2% and 37.1%), which parallels the slice example of the previous subsection and shows that the pretrained model does transfer to sentence-style queries when the geometry and the CLIP embedding of the phrase are distinctive. Plausible reasons are the distinctive geometry of the blade region and the closeness of the CLIP embeddings of the cutting phrases to that of the word “cut”.

The score maps carry more information than the arg-max reveals. For 40 objects we correlated the score map produced by the canonical word with the map produced by a question or a description. The mean Pearson correlation per affordance ranges from 0.60 to 0.89 (overall mean 0.75 for canonical against question, 0.74 for canonical against description), and the maps for a question and a description agree even more (mean 0.87). The spatial pattern of the response is therefore largely retained across phrasings, which is consistent with the

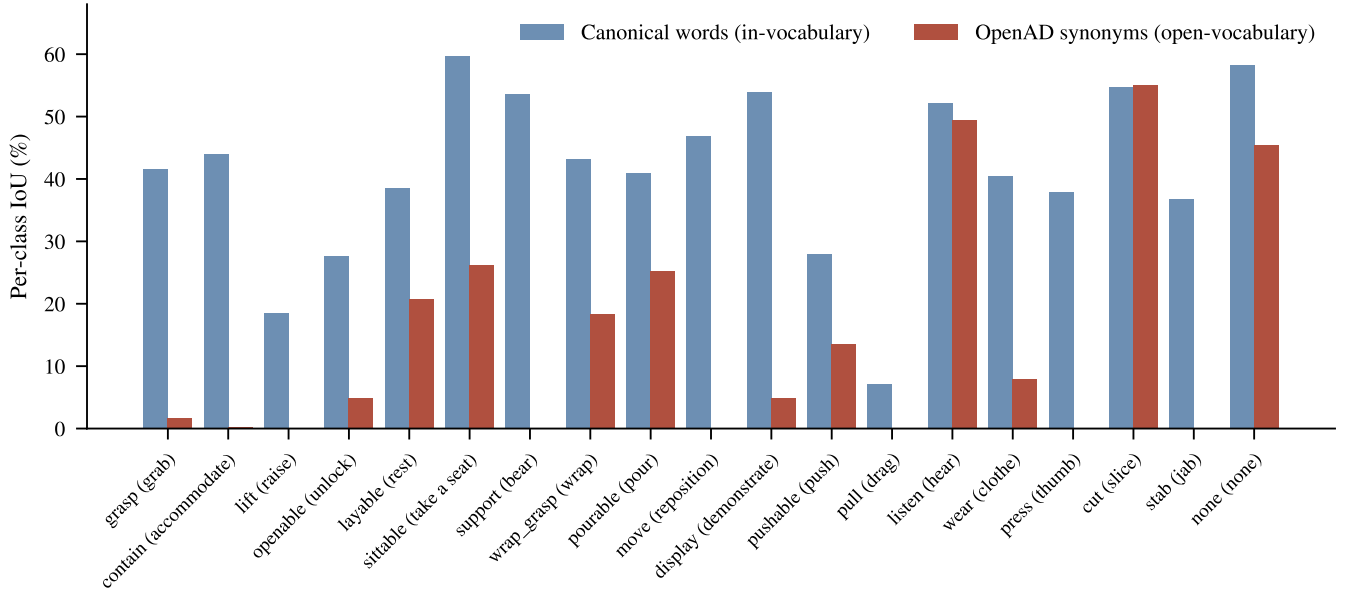


Fig. 2. Per-class IoU of the pretrained checkpoint with the canonical words and with OpenAD’s synonym queries (the synonym is given in parentheses). Ten of the 18 synonyms give an IoU below 5%, and two (slice and hear) nearly match the canonical word.

TABLE IV
IOU (%) OF THE FIVE STUDIED AFFORDANCES FOR THE PRETRAINED CHECKPOINT, WITH THE CANONICAL WORD AND WITH HELD-OUT QUESTION AND DESCRIPTION PHRASINGS. THE LAST ROW IS THE mIoU (%) OVER ALL 19 CLASSES. THIS TABLE COMES FROM THE PHRASING-SENSITIVITY RUN, SO ITS VALUES DIFFER SLIGHTLY FROM TABLE V (SECTION III-F).

Affordance	Canonical	Question	Description
grasp	42.6	0.0	0.4
contain	43.9	0.0	0.3
sittable	59.6	2.0	21.8
pourable	41.5	0.0	0.0
cut	55.2	43.2	37.1
Mean IoU of the five	48.6	9.1	11.9
mIoU, 19 classes	41.53	29.63	30.28

mechanism of Section IV: the response to a rephrased query is weaker than that of the canonical words it competes with, but it is spatially related. This result shapes the next experiment. Because the localisation is retained, the phrase-dependent part of the behaviour is expected to sit in the mapping from point features to the text space, which is exactly what prompt-augmented fine-tuning adapts. The subsample consists of the first 40 objects of the split (22 doors and 18 clocks) and serves as indicative evidence.

C. Effect of prompt-augmented fine-tuning

Table V and Fig. 3 give the main comparison for the reported model, which trains the alignment head and the last feature-propagation layer. The pretrained values are those of the same comparison run. Four observations follow, and then their implication.

Held-out phrasings improve by about ten points. On phrasings never used for training, mIoU rises from 29.27 to 40.48 for questions (+11.21) and from 30.40 to 40.88 for descriptions (+10.48). With canonical words scoring 41.28 for the pretrained model, this closes 93% and 96% of the gap that phrasing had opened. The change is more than twenty times the run-to-run variation of about 0.4 points (Section III-F), and it holds for phrasings that the model never saw in training. The head-only and decoder configurations reach the same level (+10.5 and +10.0, and +10.9 and +10.3; Section VI-E), so the result does not depend on a single configuration.

Seen and held-out phrasings behave alike. Phrases from the training set improve by +13.08 and +12.72 and reach 42.12 and 42.37, the level of the canonical words (42.02). The held-out results are only 1.6 and 1.5 points below, so the network has learned a property of the phrase family and not the 35 training strings. The held-out strings differ from the training strings as wordings and not necessarily as meanings (Section III-E), so the experiment measures generalisation across wording.

Canonical words are preserved. The canonical mIoU changes from 41.28 to 42.02 (+0.74), so robustness to phrasing is gained without a cost on the original words. The checkpoint was selected for its canonical validation mIoU on the same objects (42.46 at epoch 4, Table VI), and re-evaluating it gives 42.02, so we read the result as preservation of the original performance.

Gains are largest where the pretrained model was weakest (Fig. 3b,c). The mean IoU of the five affordances rises from 8.9% to 43.9% for held-out questions and from 12.1% to 45.0% for held-out descriptions. Averaged over the two forms, the largest gains are for sittable, contain and pourable. Grasp

TABLE V

PRETRAINED VERSUS THE REPORTED FINE-TUNED MODEL (HEAD AND LAST PROPAGATION LAYER), mIoU (%) OVER 19 CLASSES, SAME VALIDATION OBJECTS AND SAME QUERY LISTS IN ONE COMPARISON RUN. “SEEN” PHRASINGS BELONG TO THE FINE-TUNING SET, “HELD-OUT” PHRASINGS DO NOT. BOLD MARKS THE HELD-OUT CONDITIONS THAT DEFINE THE RESULT. ALL VALUES ARE OURS.

Condition	Pretrained	Fine-tuned	Δ
Canonical words	41.28	42.02	+0.74
Question, seen	29.04	42.12	+13.08
Question, held-out	29.27	40.48	+11.21
Description, seen	29.64	42.37	+12.72
Description, held-out	30.40	40.88	+10.48

under the question form has the most headroom left, with an IoU of 22.2% against about 42% for the canonical word, while the description form reaches 40.4%. For cut, where the pretrained model was already fairly stable, the gains are smaller (+11 and +16 points). We interpret only the larger per-class differences, since the noise of single classes has not been estimated.

What the result implies. Only the 101,377 parameters of the head and of the last propagation layer changed, and the capacity ladder of Section VI-E shows that the head alone and the whole decoder lead to the same result. The frozen features therefore already localise the affordances, and the improvement is an alignment of these features with sentence-style text. This is consistent with the score-map correlations of the pretrained model, and a direct probe of the features is a natural follow-up.

D. Training behaviour

Table VI lists the training record of the reported model. The loss falls from 0.701 to 0.639 and changes little after epoch 4. The canonical validation mIoU is 41.72 after the first epoch, at the level of the pretrained evaluations (41.25 to 41.53), reaches 42.46 at epoch 4 and stays between 41.98 and 42.23 afterwards, so the adaptation is fast and stable and preserves the original vocabulary. Epochs after the fourth are equivalent to epoch 4 within the evaluation noise. The head-only configuration behaves alike: its loss falls from 0.726 to 0.645 and its validation mIoU peaks at 42.22 at epoch 4 of 5. All three configurations of the capacity ladder select epoch 4 (Table VIII), so about four passes over the training set suffice at every capacity.

E. Ablations

The design of the conditions gives three factor comparisons, namely phrase form (question or description), exposure (seen or held-out) and model (pretrained or fine-tuned), and the capacity ladder adds a fourth factor. Form has little effect on either model (differences of 0.2 to 1.0 points in Table V).

How much of the network to train. The capacity ladder trains the alignment head (67,073 parameters, 5 epochs), the head together with `fp1` (101,377 parameters, 8 epochs) and the head together with the whole feature-propagation decoder

TABLE VI

FINE-TUNING RECORD OF THE REPORTED MODEL (HEAD AND LAST PROPAGATION LAYER). LOSS IS THE MEAN WEIGHTED NLL OF THE EPOCH. VALIDATION mIoU USES CANONICAL WORDS ONLY AND IS THE CRITERION FOR CHOOSING THE CHECKPOINT.

Epoch	Train loss	Val. mIoU (%)	Time (s)
1	0.701	41.72	880
2	0.650	42.05	878
3	0.645	42.18	872
4	0.644	42.46	856
5	0.642	42.23	844
6	0.641	41.98	830
7	0.640	42.17	835
8	0.639	42.15	839

(743,041 parameters, 8 epochs), with the same data, phrases, loss, learning rate and seed. The rungs differ by a factor of eleven in trainable parameters, and each is compared with the pretrained model of its own comparison run (Table VII). The gain is present at every rung. Head-only training already delivers +10.5 and +10.0 mIoU on held-out questions and descriptions and +0.5 on canonical words. Adding `fp1` raises the held-out scores by a further 0.73 and 0.61 points, to 40.48 and 40.88, the highest values of the ladder. Releasing the whole decoder gives 40.20 and 40.51, within 0.3 to 0.4 points of the previous rung and inside the range of repeated evaluations (0.41 points). The per-affordance IoUs of the head-only and head plus `fp1` models differ by less than 3 points in eight of the ten affordance and form combinations (the exceptions are grasp and cut under questions, +7.4 and +4.1 points). The ladder supports two conclusions. First, the alignment head carries nearly all of the gain, so the update that is needed is small. Second, the improvement is stable over a factor of eleven in capacity, which indicates that it is a property of the paraphrase supervision and not of a particular parameter budget. We report head plus `fp1` as the main model because it has the highest held-out scores and the highest canonical validation mIoU (Table VIII) at a small extra cost over the head alone.

VII. QUALITATIVE ANALYSIS, FAILURE CASES AND DEMONSTRATION

A. Examples

For six pairs of object category and affordance we render the ground-truth mask, the pretrained score map and the fine-tuned score map under a held-out question (Figs. 4 and 5). The fine-tuned model in these figures is the reported one, which trains the alignment head and the last propagation layer. For each category we used the first validation object whose ground truth contains the affordance, so the object was not selected by looking at model output. The category and affordance pairs were chosen by us to include frequent and rare cases. All panels share one colour scale. The maps show the probability of the query against the single competitor `none`, not the 19-way arg-max used for mIoU, so they show relative confidence and cannot be read as the segmentations that Section VI scores.

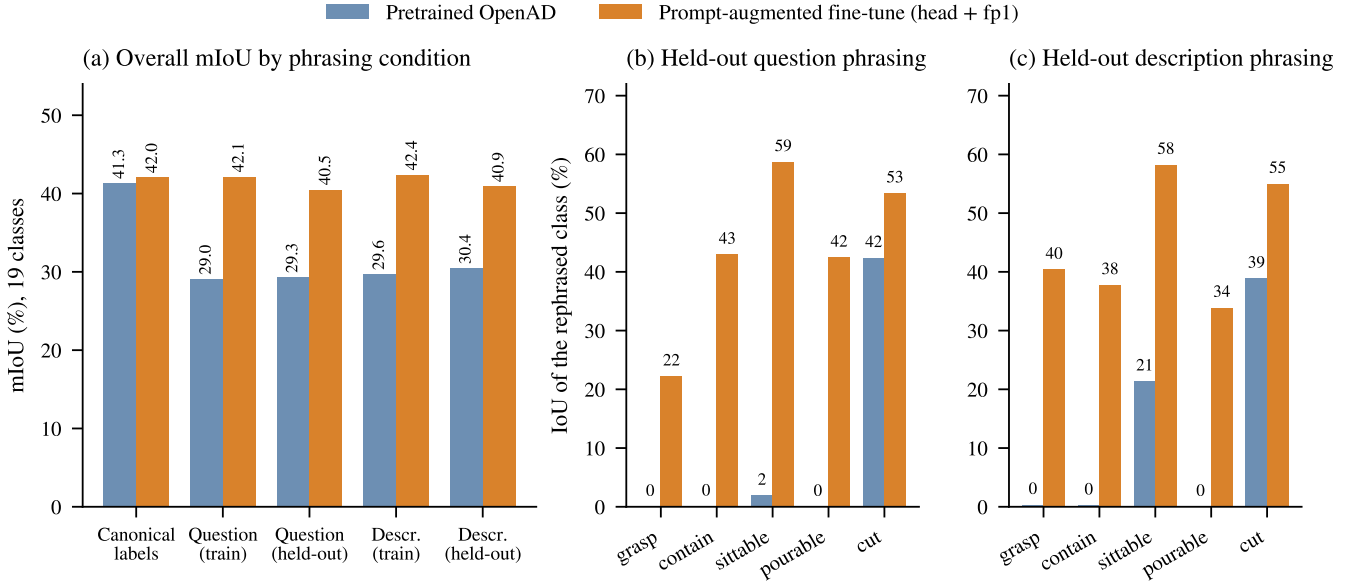


Fig. 3. (a) mIoU of the pretrained and the reported fine-tuned model (head and last propagation layer) under the five phrasing conditions of Table V. (b, c) IoU of each reprased affordance under held-out question and description phrasings. Grasp with the held-out question remains the weakest case after fine-tuning.

TABLE VII

CAPACITY LADDER, mIoU (%) OVER 19 CLASSES. EACH FINE-TUNED MODEL IS COMPARED WITH THE PRETRAINED MODEL OF ITS OWN COMPARISON RUN, SO THE PRETRAINED ROWS DIFFER SLIGHTLY (SECTION III-F). THE SELECTED CHECKPOINT IS FROM EPOCH 4 IN ALL THREE CONFIGURATIONS.

Model	Params trained	Canon. words	Question held-out	Descr. held-out
Pretrained, run 1	n/a	41.44	29.22	30.22
Head only	67,073	41.93	39.75	40.27
change vs run 1		+0.49	+10.53	+10.04
Pretrained, run 2	n/a	41.28	29.27	30.40
Head + fp1 (reported)	101,377	42.02	40.48	40.88
change vs run 2		+0.74	+11.21	+10.48
Pretrained, run 3	n/a	41.19	29.29	30.23
Decoder (fp1-fp3)	743,041	41.73	40.20	40.51
change vs run 3		+0.54	+10.91	+10.28

The six objects illustrate the behaviour that we observed and are not a statistical sample.

Clear improvement (chair, bowl). For the chair, the pretrained model gives intermediate scores over the annotated region and its surroundings, and the fine-tuned model gives scores near 1 over most of the annotated region with a clear boundary, apart from a weaker part at its lower edge. For the bowl, the pretrained map is uniformly low to moderate, while the fine-tuned map is high on the interior surface that carries the label and moderate on the outer shell, so the region is found and its extent is slightly overestimated.

Partial improvement (knife). Both models peak at the same small region of the blade edge, and the fine-tuned map has a

TABLE VIII

FINE-TUNING RECORDS OF THE THREE CONFIGURATIONS OF THE CAPACITY LADDER (“Loss” IS THE TRAINING LOSS OF THE LAST EPOCH). VALIDATION mIoU USES CANONICAL WORDS, IS THE SELECTION CRITERION AND IS THE BEST EPOCH’S VALUE (EPOCH 4 IN ALL THREE). TIME IS THE TOTAL FOR ALL EPOCHS, EACH INCLUDING A FULL VALIDATION PASS.

Config.	Params	Ep.	Loss	Val. (%)	Time (s)
Head only	67,073	5	0.645	42.22	4,261
Head + fp1	101,377	8	0.639	42.46	6,834
Decoder	743,041	8	0.656	42.07	6,726

stronger peak. The rest of the object receives a raised score of about 0.4 to 0.6, which agrees with the smaller aggregate gain for cut, where the pretrained model was already fairly stable.

Broad activation (mug, bottle). For the mug, the fine-tuned map raises a near-zero pretrained response to above 0.6 over most of the object, so the object is detected as pourable while the labelled rim region is not yet delimited. For the bottle, the fine-tuned model produces a peak at the labelled patch on a background of 0.3 to 0.5, where the pretrained model produces nothing.

Failure case (bag). For the bag, neither model gives a score above the background level on the labelled grasp region. The same affordance and question form are detected on the bottle, and the bag agrees with the lower aggregate IoU of grasp under held-out questions (22.2% for the reported model). Plausible contributing factors are that the validation split contains only 12 bags, that the grasp region on bags (handles and body) is visually diverse, and that the sentence “Where should I take hold of this?” may lie far from the seven training phrases of

grasp in the text space. A larger set of training phrases for grasp is a direct way to test the last factor.

Across the examples, fine-tuning raises the score of the queried class over broad parts of the object, which gives high recall, and the maps are sharpest where the labelled region is large (chair, bowl). Whether the wider maps on small regions (mug, knife, bottle) reflect a change in spatial precision, or only a shift of the scores of the whole object that would leave the ranking of the points intact, can be settled by a threshold-free measure such as the area under the ROC curve per object. We identify this as the natural next analysis.

B. Errors that do not depend on phrasing

Part of the remaining error does not depend on the wording. Rare or ambiguous affordances have low IoU with canonical words in Fig. 2, namely pull (7.1%), lift (18.5%), openable (27.7%) and pushable (28.0%). Per-object point accuracy with canonical words is lowest for Bag (41% averaged over 12 objects), Bottle (44%, 41 objects) and Hat (46%, 22 objects), and highest for Keyboard (92%, 15 objects) and Display (84%, 88 objects). These errors are properties of the underlying detector and the data. They mark the part of the problem that lies outside phrasing robustness, which changes only the alignment head and one propagation layer.

C. Interactive prototype

The same model is exposed through an interface built with Gradio and Plotly, so that the findings above can be explored interactively. The user selects an object from the validation split and a model (pretrained or fine-tuned), types a query or picks an example, and receives a rotatable 3D point cloud coloured by the score of Eq. (2) against `none`, together with the fraction of points with a score of at least 0.5 and the peak and mean score. Predictions are cached per object, query and model, so repeated queries are immediate. The interface distinguishes four states, namely a normal result, a low-confidence result (no point reaches 0.5, flagged as unreliable), an empty query, and an inference error. It also keeps the previous query for side-by-side comparison, which lets a user see directly how the wording changes the map. These features follow from the results above: because phrasing changes the maps, comparing two wordings is useful, and because a uniformly low map can mean either an absent affordance or a phrasing failure, the interface states its uncertainty explicitly. The logic of the backend is unit tested. Usability is examined in a separate report on the human-computer interaction aspects of the project.

VIII. SCOPE AND EXTENSIONS

Statistics. Every number comes from one evaluation pass and one fine-tuning run per configuration (seed 1). The harness resolves differences of about 0.4 mIoU at the level of the total (Section III-F), and we interpret differences above this level and only large per-class differences. The same effect in three separately trained configurations supports the main result.

Data. The distributed data contain a training and a validation split and no separate test split. The validation split serves to choose the epoch, by the canonical mIoU alone, and for the final evaluation. Held-out phrasings play no role in the choice of the epoch. The choice of the reported configuration among the three rungs of the ladder used the held-out scores, and all three rungs are therefore reported.

Phrasing. The study covers five of the 18 affordances, each with one held-out question and one held-out description (ten strings), which is a controlled probe of generalisation across wording. The held-out strings are disjoint from the training strings, and their semantic distance from them is not controlled. Broader phrase sets and the held-out label forms, which are defined in the configuration, extend the same protocol.

Interpretation. The score-map analysis used 40 objects, all doors or clocks, and gives indicative evidence. The account of why phrasing matters (Section IV) and why a small update suffices follows from aggregate metrics, and an analysis of the embedding space is a natural extension.

Extensions. The protocol extends directly to a canonical-only fine-tuning control (the `--no-augment` option of our script), to an open-vocabulary (synonym) evaluation of the fine-tuned models, to fine-tuning of the full network including the encoder, and to further seeds. These are the first of the next steps listed in Section IX.

IX. CONCLUSION

We reproduced the open-vocabulary result of OpenAD with the released checkpoint (14.40 against 14.37 mIoU), built an evaluation harness that scores arbitrary query strings under identical conditions, and placed its scores on the metric scale with two query-independent floors (2.27 and 1.47 mIoU). With the harness we showed that the wording of the query moves the mIoU of the released model by about 12 points when five affordances are phrased naturally, that the change is concentrated in the rephrased classes, and that the score maps stay spatially correlated across phrasings. These findings pointed to the alignment between point features and sentence-style text. Prompt-augmented fine-tuning of the alignment head and the last propagation layer, only 101,377 parameters, recovers 93% and 96% of the difference on held-out question and description phrasings (+11.2 and +10.5 mIoU) and preserves the score of the original words. A capacity ladder of three configurations, from 67,073 to 743,041 parameters, gives the same gain at every rung (+10.0 to +11.2 mIoU on held-out phrasings), which places the improvement in the alignment head and shows that it does not depend on the parameter budget. The qualitative study shows the recovered regions and the cases with remaining headroom, for example grasping a bag, and an interactive prototype makes the effect of wording visible to users. The study covers five affordances, ten held-out strings and a single seed. The next steps are a canonical-only control, an open-vocabulary evaluation of the fine-tuned models, more phrases per affordance, several seeds, and a training objective

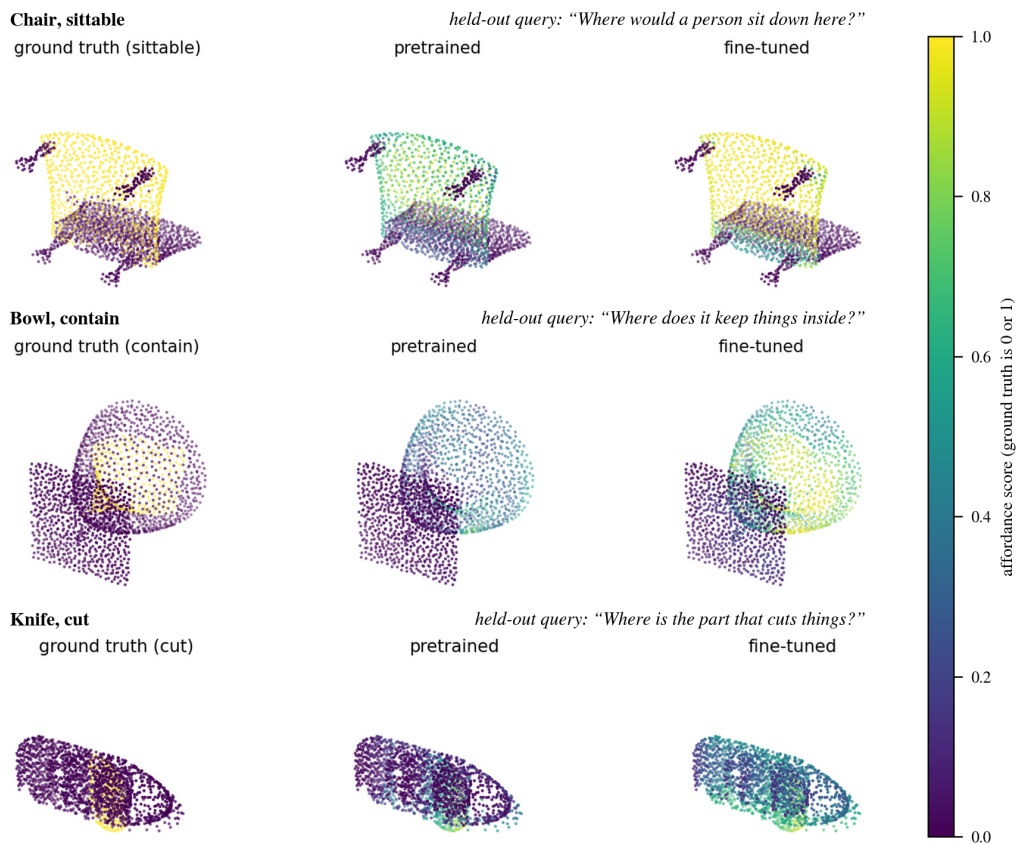


Fig. 4. Improvement cases with held-out question queries. From left to right the panels show the ground truth (0 or 1), pretrained score, score of the reported fine-tuned model, on the same colour scale. The fine-tuned model recovers the region for the chair and the bowl, and the knife blade shows a stronger peak with a raised background.

that penalises disagreement between the maps of paraphrases directly.

Availability. Code, frozen prompt sets and the commands that produce every table are in the repository cited in the introduction. The figures of this report are generated from the stored metric files by a script in the same repository. The data and the pretrained checkpoint are those released by the OpenAD authors and are not redistributed. OpenAD is released under the MIT licence.

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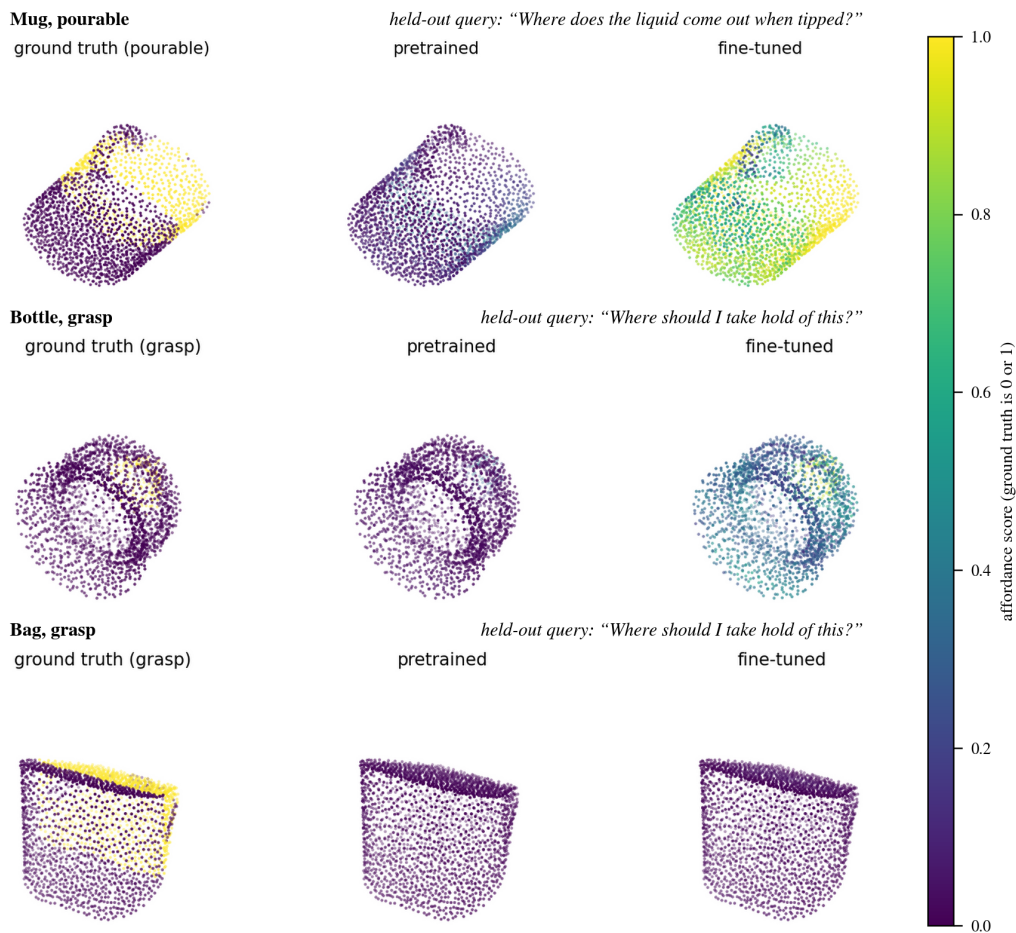


Fig. 5. Further cases of the reported fine-tuned model with held-out question queries. The mug shows a broad activation, the bottle a correct peak on a raised background, and the bag is a failure case of both models.

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